Estimating the Influence of Demographic Ageing and Migration to the Spanish Aggregate Labour Supply

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Abstract

Population aging has raised concerns about its impact on the labor supply. However, quantifying these effects often focuses only on the increase in the elderly population, at the expense of the working age groups. This article estimates the aggregate labor supply in Spain and examines the extent to which population aging, immigration, and individual behaviors have contributed to its growth between 2006 and 2023.

Keywords: Labour Supply, Immigration, Population Ageing, Decomposition Models.

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Contents

1 Introduction **3**

2 Litterature Review **3**

3 Data Source and Methodology **6**

3.1 Measuring the Aggregate Labor Supply 6

3.2 Data sources and descriptives 6

3.3 The Model of Continuous Change 6

4 Results and Analysis **8**

5 Discussion and Conclusion **10**

1 Introduction

Spain offers a timely and policy-relevant context for studying how demographic ageing, individual behavioural change, and immigration jointly shape aggregate labour supply. Ageing is reshaping the population's age structure, with direct implications for labour-force participation and hours worked. At the same time, sustained immigration is changing the composition of the working-age population and influencing labour-supply dynamics. Because these processes operate through overlapping demographic and economic channels, analysing them separately can produce incomplete or misleading conclusions.

This article addresses that gap through an integrated framework that links age structure, participation behaviour, hours worked, and migration-driven population change. It makes two main contributions. First, it provides a more precise measure of aggregate labour supply by combining participation rates and hours worked by age group and residence status for the population aged 15 and over. Second, it isolates the contribution of demographic ageing by decomposing changes in aggregate labour supply into demographic and behavioural components.

This approach is essential for identifying whether observed labour-supply changes are driven mainly by structural demographic shifts, behavioural responses in participation and work intensity, or immigration inflows. By separating these channels, the analysis moves beyond broad claims about "ageing" or "migration" and provides evidence that is more useful for policy design. Although isolating the effects of demographic ageing on aggregate labour supply is methodologically challenging, it is necessary for clearer policy discussion and better-targeted interventions. The next section presents the methodology and data.

Graphics

2 Litterature Review

Across the countries of Western Europe, Spain is a representative case that exhibits two major phenomena that currently concern policymakers: population ageing and immigrant inflows. These conditions result in several economic and social outcomes. From an initial standpoint, it is widely recognized that there are side-effects to an ageing society, which are evident in social security finances under substantial stress affecting PAYG systems (Díaz-Giménez and Díaz-Saavedra, 2009; Boeri et al., 2024), increments in public expenditure on health services (Cristea et al., 2020; Aiyar and Ebeke, 2016; Bijak et al., 2007), and social security (Faílde Garrido et al., 2021). Furthermore, the ageing population impacts the labour market as there exist a decline in labour force (Maestas et al., 2023). Concerning the labour market in Spain, Casares and Vázquez (2018), demonstrated that labour force shocks are more persistent, and this labour force growth can be explained by separate factors and one such factor is the influx of a substantial number of non-EU immigrants. This suggest that immigration influences the Spanish aggregate labour supply besides the age structure of the population.

As a result, several investigations have suggested that replacement migration is proposed as a strategy to alleviate the effects of ageing because it assists in easing downturns in labour supply, boost competitiveness and fosters economic growth (Stepanek, 2022; Okamoto, 2021; UN, 2000). However, the expansion of working population and, therefore, the increase in public revenues, migration also creates additional needs for public services. Consequently, migration affects both the tax and the welfare system (Fiorio et al., 2023; Naumann and Hess, 2021). For example, as stated in Preston (2014), immigration can result in costs by the utilization of public services but can also influence the expense of offering services to natives. Having said that, in developing countries, extended individual longevity or population ageing and migration are among the most significant challenges (Ciobanu et al., 2020).

Spain, as various developed economies, present population ageing owing to the increasing life expectancy and falling fertility rates. This demographic trend impacts the job markets and the influence of the reduction of labour supply gets more evident as the "baby boomer" generation is getting older and ready for retirement (Casares and Vázquez, 2018). The present circumstances in the Spanish labour market are a result of intense and continous unemployment rates which have been explained through imbalances between labour supply and demand. Several facts shape the Spanish labour market conditions among which the institutional framework (types of employment contracts), demographic composition of the population, economic sensitivity to fluctuations, the grey economy, and internal migration related to the redistribution of human capital between natives and immigrants (Bentolila and Dolado, 1990; Bentolila et al., 2010; Congregado et al., 2014; Güell et al., 2021). Overall, unemployment in Spain is seen as a structural challenge (Bentolila et al., 2020). Meanwhile the contributions in terms of individual participation has experienced different phases throughout the years. This highlights that in both economic growth and contraction, the way different types of individuals respond to economic activity is essential to understanding the compostition of labour supply.

The Spanish economy has presented increasing rates of economic activity during 2004 to 2007, commonly this period is understood as an expansionary phase. However, as many other economies, during the economic crisis of 2008, this upward trend has moved in the opposite way. In the phase before the Great Recession, the demographic profile of unemployment in Spain was shaped by an average total unemployment rate of 8,2% in 2007, the lowest unemployment rate achieved since becoming part of the European Union (EU) in 1986. This population characteristics within the job market status showed that the youth unemployment rate remained low near the values captured of total unemployment, albeit high in comparison to other EU countries. Moreover, female conditions in the Spanish labour market, had also attained the lowest unemployment rate reached, achieving an average value of 10,7% in 2007. Simultaneously, during the pre-crisis growth period, a large wave of immigrants arrived in Spain altered the demographic structure of the working population (Amuedo-Dorantes and De La Rica, 2008; Arango, 2013; Gonzalez and Ortega, 2011). Nonetheless, the international

financial downturn alongside substantial housing and mortgage crisis caused the economy of Spain to experience considerable impacts regarding individual's behaviour in the job market.

As stated before, the high volatility of unemployment during economic cycles impact people's engagement in the labour force and their retirement decisions. Within a framework where the economy deals with population ageing, the Spanish labour market participation has been explained in multiple manners. From one side, labour supply by gender has taken importance due to the deep impact of the Great Recession on the Spanish labour market. In terms of decisions to join the working population during times of crisis, most of the results from analysis associated to gender-based workforce supply patterns emphasize that the presence of a countercyclical behaviour of labour supply by gender takes importance in Spain. This conduct is referred to as added-worker effect (AWE), where the focus is on the increase of the involvement of women in the active population as the couple falls into unemployment and, to a lower extent, when the partner is self-employed. Meaning that AWE influences women's decision-making during economic crises, particularly in countries that are highly responsive to economic cycles, such as Spain (Congregado et al., 2014; Sanz-de Galdeano and Terskaya, 2020). Conversely, immigrant employment declined leading to a decrease of foreign workforce and working-age individuals (Rodríguez-Planas and Nollenberger, 2016). Consequently, these labour participation patterns across the two phases of the recession, the international financial crisis and the eurozone crisis, changed the structure of labour supply. In other words, during the recession in Spain, most of the labour force participation was attributed by the increase of female workers in the service sector, a male non-dominated economic sector, and a drop in the share of immigrants in the workforce.

The migration process has taken importance in the early 2000s, when Spain used to be considered as a source of emigration. Over the past years, the increasing inflows experienced shaped the structure of the working population. This trend was hit during the economic downturn and reduced foreign migrations in Spain. According to the National Institute of Statistics of Spain (Instituto Nacional de Estadística, INE), the country registered 599,074 immigrations in 2008, the number of immigrants peaked during this year. Therefore, the share of immigrants in the working-age population was around 13,9%. However, the burden of job losses during the crisis was disproportionately endured by foreign workers, by 2013 the share of immigrants in the labour market reached in average 11%. The immigration inflow from abroad fell more than 50% between 2008 and 2013. Consequently, Spain faced net migration outflows. In contrast, after the turbulences faced in this country, the evolution of the immigration inflows recovered the initial behaviour. In 2014, the rate of immigrants entering the country was 8,8%, while in 2019 larger increments of internal migration flows raised. Regardless of the health crisis of the COVID-19, the ongoing influx of foreign arrivals kept increasing resulting in 1,258,894 immigrants in 2022, making the largest number of immigrants to date.

3 Data Source and Methodology

3.1 Measuring the Aggregate Labor Supply

Aggregate labor supply L is estimated by the weighted sum of hours worked during the year by age a and by residence status r (whx_{ar}), with $r =$ (immigrant, native).

The weighting factor is the product of the total population size (pop_r), the population structure (str_{ar}), and the labor participation rates (wpx_{ar}) by age and residence status.

$$L = \sum_r \sum_a (pop_r \times str_{ar} \times wpx_{ar} \times whx_{ar}) = f(pop_r, str_{ar}, wpx_{ar}, whx_{ar}) \tag{1}$$

3.2 Data sources and descriptives

3.3 The Model of Continuous Change

To measure the isolated effect of demographic aging on the aggregate labor supply, the continuous change model (Horiuchi et al., 2008) was used. This model allows decomposing the difference between two summary measures resulting from the same process into several components, each representing the contribution of underlying factors to the process. The process is a function f, taking as arguments the values of the factors (covariates or independent variables) and returning a summary measure (the dependent variable or variable of interest).

Horiuchi et al. (2008) demonstrates that as the factors transition from values $x_i(t_1)$ to $x_i(t_2)$ between two times t_1 and t_2, the summary measure changes from $f(x_i(t_1))$ to $f(x_i(t_2))$, and the difference between these two measures can be decomposed into additive components c_i representing the contribution of changes within each factor to this difference. This is expressed by equation (2).

$$f(x_i(t_2)) - f(x_i(t_1)) = \sum_i c_i \tag{2}$$

With $i = 1, 2, ..., n$ being the factors involved. The decomposition is based on the assumption that the change in the covariate x_i occurs continuously or gradually over a real or hypothetical time dimension $t_1 \rightarrow t_2$. This provides a reasonable justification for the additivity of the effects of the factors and the elimination of interaction terms even if the process in question is a non-additive function of the covariates (Horiuchi et al., 2008, p. 786).

In other words, the continuous change model (CCM) introduces a decomposition function g which, using a process function f, transforms two series of covariates, the vectors $X(t_1)$ and $X(t_2)$ representing the values of the covariates $X = (x_1, ..., x_i, ..., x_n)$ at times t_1 and t_2, into a series of contributions or effects, the vector $C = (c_1, ..., c_i, ..., c_n)$. Thus, we can write:

$$C = g(X, f) \tag{3}$$

The series X and C having the same lengths, results in a perfect correspondence between the factors and the effects, which can be grouped for analysis purposes. Furthermore, the idea of continuity implies that each element c_i of the series C is obtained by integration according to the equation:

$$c_i = \int_{x_i(t_1)}^{x_i(t_2)} \frac{\partial f(t)}{\partial x_i(t)} dx_i(t) \tag{4}$$

In practice, equation (4) is estimated by numerical integration where $\int_{x_i(t_1)}^{x_i(t_2)} d(t)$ is approximated by $\sum_{x_i(t_1)}^{x_i(t_2)} \delta(t)$, with $x_{it} = x_i(t)$ being the value of a factor i at a moment t. Adapting the continuous change model to measure equation (1) allowed the total variation in the aggregate labor supply between 1981 and 2016 to be distributed not only among these three components but also by age group and residence status (immigrants and natives). For this article, the decomposition was performed by creating an auxiliary function g around the R package (R Core Team, 2018) DemoDecomp (Riffe, 2018). The code and data used are available upon request.

The CCM model is a better alternative to the standardization method often used in demography to account for differences in population structure when comparing demographic measures. Indeed, standardization requires a third population to serve as a standard, making it somewhat less robust as different standards can lead to different results. The CCM model solves this problem and allows for direct decomposition without interaction terms between the different factors.

Interaction effects in regression analysis refer to situations where the impact of one variable on a dependent variable is not constant or additive but depends on the values of other variables. In traditional decomposition methods, such as standardization, any discrepancy between the overall change and the sum of the individual effects of the variables is also called an interaction effect. In such cases, the interaction represents an incomplete separation of the contributions of individual covariates to the overall change or difference in a dependent variable.

4 Results and Analysis

Figure 1: Average participation rate by major age groups for immigrants and natives, Spain 2006-2023

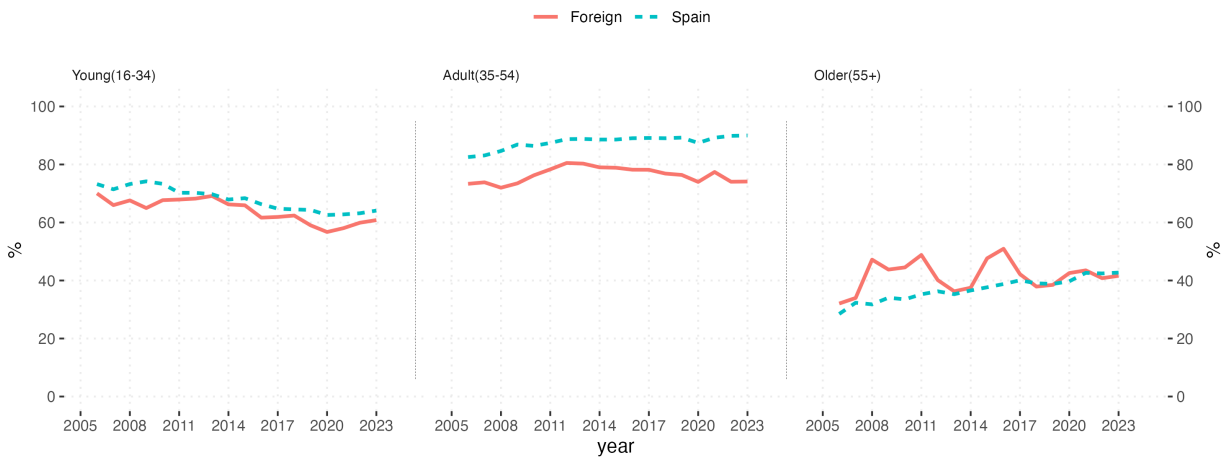


Figure 2: Average worked hours by major age groups for immigrants and natives, Spain 2006-2023

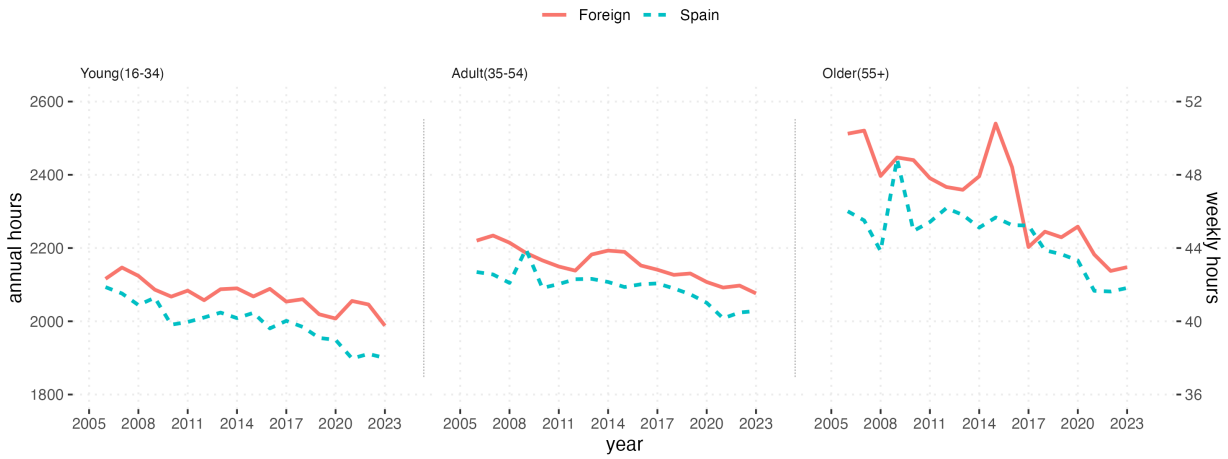
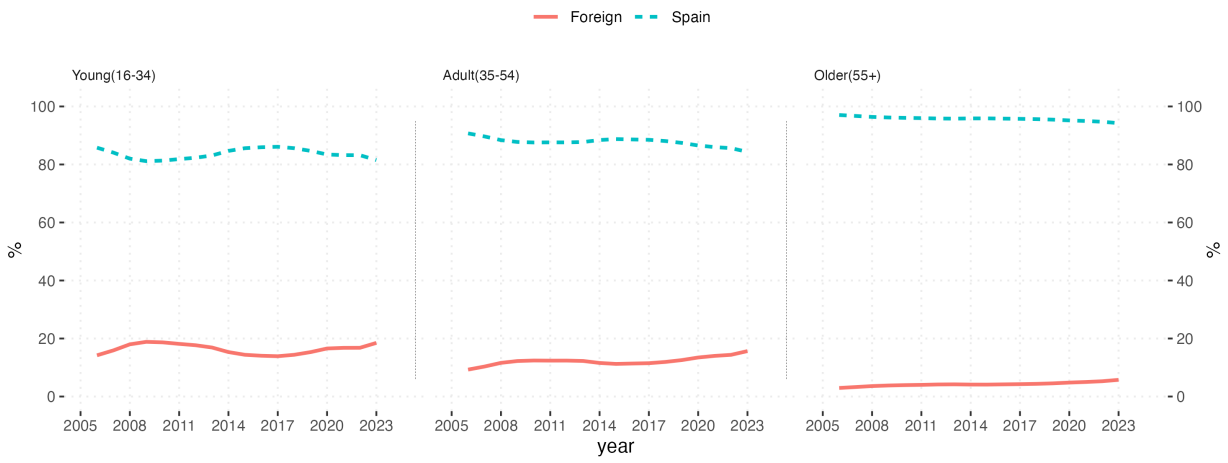


Figure 3: Percentage of total population by major age groups for immigrants and natives, Spain 2006-2023



5 Discussion and Conclusion

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